Voyage Analytics : Production-Grade MLOps Travel Platform

A production-grade, end-to-end Machine Learning Operations (MLOps) travel platform designed to demonstrate the complete lifecycle of predictive models in a live environment. It combines three different machine learning systems into one travel intelligence application : flight price prediction, traveler profiling, and hotel recommendation. Along with the models, we implemented data validation, feature engineering, experiment tracking, API serving, monitoring, testing, containerization and deployment.

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Course : Labmentix Major Project

PROBLEM STATEMENT

The Core Challenges in Travel Tech & MLOps :

- Modern travel platforms generate large amounts of data from **flights, hotels, bookings and traveler behavior**, but these data sources are often disconnected.
- **Isolated Travel Data :-** Flight information, hotel bookings and traveler profiles exist in separate datasets.This makes it difficult to create a unified view of a traveler's journey.Traditional systems often treat these datasets independently instead of combining them for personalization.
- **Flight Price Uncertainty :-** vary according to Route, Travel class, Airline/agency, Travel duration, Season/month, Day of the week & Holidays. Travelers therefore have difficulty estimating whether a current ticket price is reasonable.
- **Limited Traveler Understanding :-** Demographic information may be incomplete or noisy. Behavioral travel information can contain useful signals about a traveler. A machine learning model can use behavioral features to infer traveler characteristics and create meaningful traveler personas.
- **Hotel Recommendation & Cold Start :-** Generic hotel recommendations do not consider individual traveler preferences. Collaborative filtering works well when historical user-rating information exists. However, a new user has no previous ratings, creating the cold-start problem.
- **ML Models Alone Are Not Enough :-** A production system needs Data validation, Reproducible preprocessing, Feature engineering, Experiment tracking, Model management, API serving, Automated testing, Deployment, Monitoring & Drift detection. The project addresses these problems through a unified **MLOps architecture**.

The main problem is not simply predicting something. The bigger problem is building a complete system where different travel intelligence tasks can work together reliably. We therefore designed the project as an MLOps platform rather than just a notebook-based machine learning project.

PROJECT OVERVIEW

Voyage Analytics 2.0 is an **end-to-end production-oriented travel intelligence platform** that combines three ML engines with a complete MLOps lifecycle.

- **Three Core ML Engines :-**

1. Flight Price Prediction : Predicts the expected price of a flight based on travel and route-related features.

Output : Predicted ticket price , 95% prediction interval , Feature contribution/explanation using SHAP

2. Traveler Profiling : Uses behavioral and transactional travel information to classify traveler demographic information and generate traveler persona insights.

Example Personas : Premium Corporate Traveler and Occasional Leisure Traveler

3. Hotel Recommendation : Provides personalized hotel recommendations using Collaborative filtering , SVD matrix factorization and Content-based similarity. The content-based approach provides a fallback for new users.

- **End-to-End Objective :-**

The platform connects : Data → ML Models → API → Dashboard → Monitoring

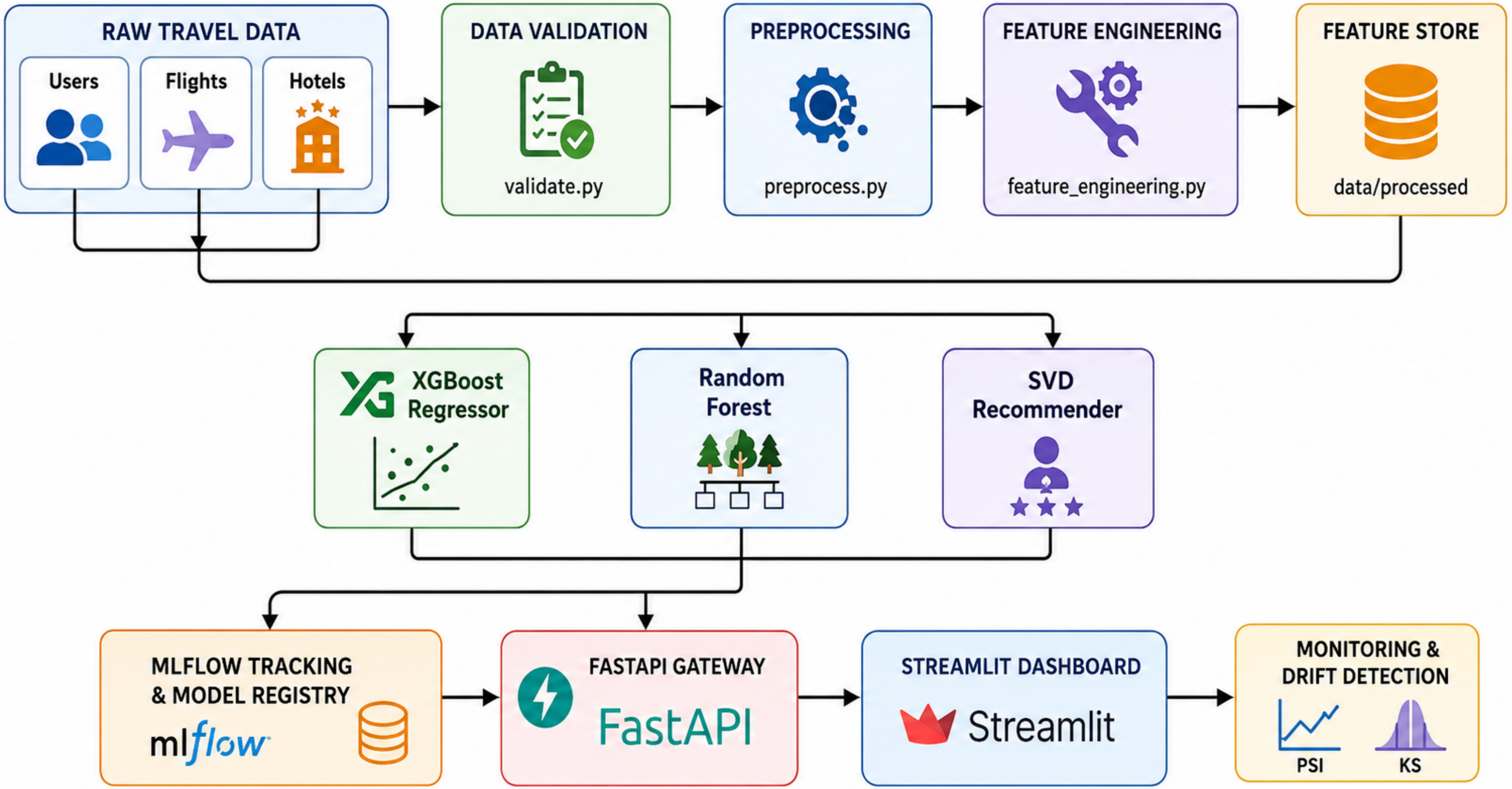
Instead of having isolated ML notebooks, the system provides a complete lifecycle : Collect → Validate → Transform → Engineer Features → Train → Track → Register → Serve → Monitor

- **Major Goals :-**
  - Build reproducible ML workflows
  - Prevent data-quality problems
  - Compare multiple algorithms
  - Tune the best-performing models
  - Deploy models through APIs
  - Provide an interactive dashboard
  - Monitor production behavior
  - Detect model-input drift

TECH STACK

Category	Technologies / Tools	Purpose / Usage
Programming & Data Processing	Python 3.10+ / Production aligned to Python 3.12, Pandas, NumPy, SciPy	Data manipulation, cleaning, transformation, numerical operations, and feature engineering
Machine Learning	Scikit-learn, XGBoost, LightGBM, Scikit-Surprise	Regression, classification, recommendation, cross-validation, and hyperparameter optimization
Explainable AI	SHAP, Plotly	Explain how individual features influence model predictions and visualize model behavior
Backend / API	FastAPI, Uvicorn, Pydantic	Create prediction endpoints, validate API payloads, serve trained models, and provide interactive API documentation
Frontend	Streamlit, Plotly	Build the interactive travel intelligence dashboard and data visualizations
MLOps	MLflow, SQLite Tracking Backend, Feature Store, Prometheus, PSI, KS Test	Experiment tracking, model management, feature management, monitoring, and data/model drift detection
Deployment & Infrastructure	Docker, Docker Compose, Kubernetes, GitHub Actions, GitHub Container Registry, Render / Streamlit Deployment	Containerization, orchestration, CI/CD, container image management, and cloud/application deployment

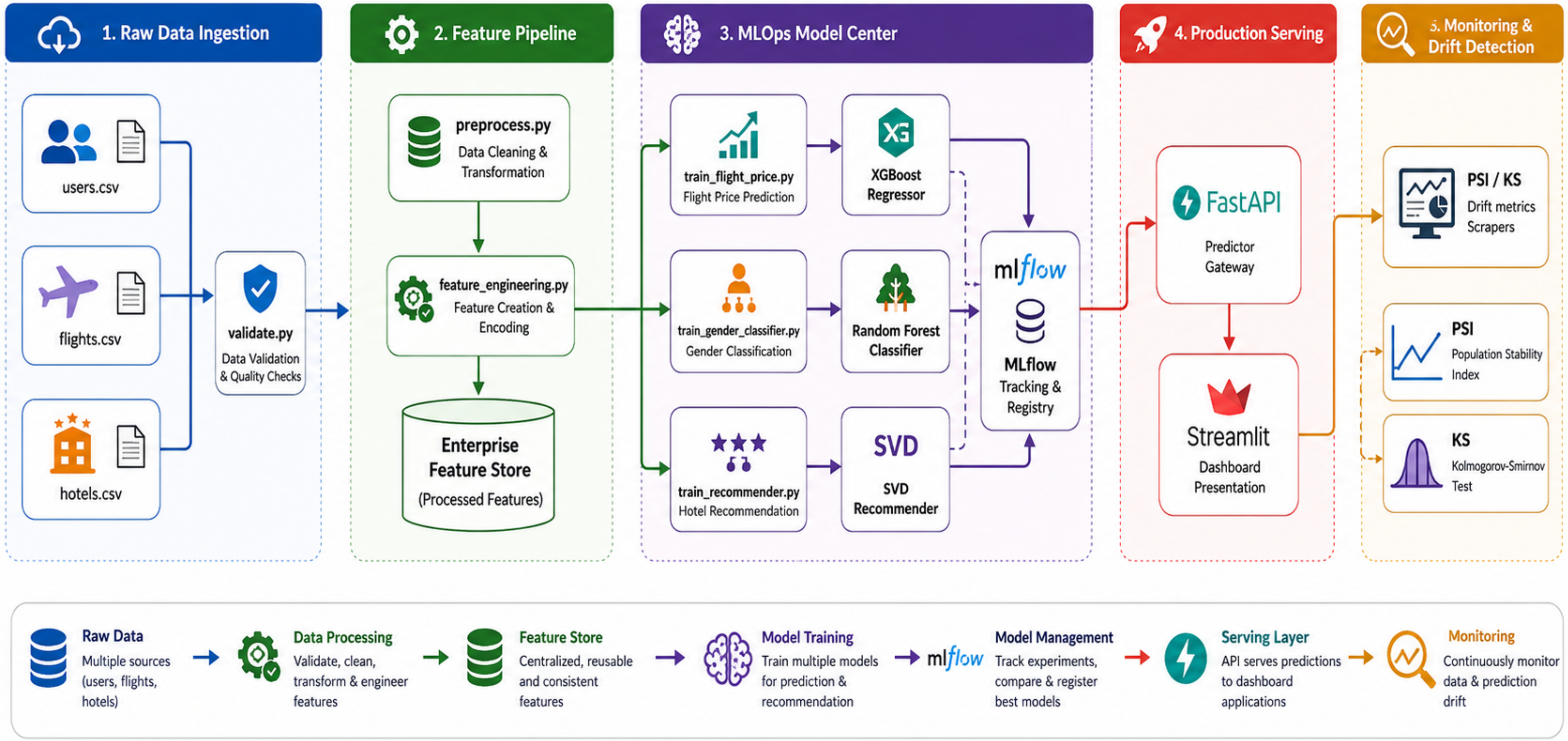
ML PIPELINE ARCHITECTURE



ML PIPELINE ARCHITECTURE

- **Data Ingestion & Validation :-** Reads transaction CSV logs of Raw datasets (users.csv, flights.csv , hotels.csv),
  validate.py checks : Schema, Data types, Missing values, Value ranges, Expected columns, Data consistency
  Checks for missing columns, enforces data types, and verifies bounds (e.g. price > 0).
- **Data Preprocessing :-** *preprocess.py* handles missing values, Cleans data, Removes duplicates, Converts categorical information, Prepares consistent training data
- **Feature Engineering :-** Extracts route distances and duration matrices from city pairs. Processes dates to extract features like departure month, weekdays, and holiday flags. *feature_engineering.py* derives useful features such as Engagement level, Delay ratio, Absolute delay, Click intensity, Activity score
- **Feature Store :-** Processed and engineered features are stored in *data/processed/* and reused by the training pipelines.
- **Model Training & Training :-** Competes multiple algorithms in parallel during training. Evaluates them on stratified test data using MLflow Tracking. Three independent training pipelines : Flight → XGBoost , Gender → Random Forest , Hotel → SVD + Content-Based
- **MLflow :-** Tracks Parameters, Metrics, Experiments, Model artifacts, Model versions
- **Production Serving :-** FastAPI loads production models and exposes prediction endpoints.
- **Dashboard :-** Streamlit provides the interactive interface.
- **Monitoring :-** Incoming data is monitored using PSI, KS statistics, API latency, Request metrics, Status codes
- **Promotion Registry :-** Promotes the candidate model with the best score (F1/RMSE) to 'Production' stage, registers metadata, and serializes the model to joblib.
- **Inference Cycle :-** Deployed FastAPI endpoints fetch the registered model from cache and serve live client prediction requests.

SYSTEM WORKFLOW



Model 1: Flight Price Prediction (Supervised Regression)

Predict the expected domestic flight ticket price in **BRL (Brazilian Real)** based on travel and route characteristics.

Input Features :- Travel class, Travel agency, Route information. Route distance, Flight duration, Month of departure, Day of week, Public holiday information, Agency scale factor

Final Model :- Tuned XGBoost Regressor (Hyperparameter optimization was performed using **5-fold RandomizedSearchCV**.)

Final Reported Parameters : learning_rate = 0.08 , max_depth = 6 , n_estimators = 100 , subsample = 0.8 ,

colsample_bytree = 0.8

Final Performance , R^2 = 0.9973

Algorithm	Baseline R^2
Linear Regression	0.5623
Ridge Regression	0.5619
Random Forest	0.9971
LightGBM	0.9961
XGBoost	0.9948

Additional project output : Approximate error variance: **±19.3 BRL &** 95% prediction interval

Explainability SHAP is used to explain individual predictions.

For example, SHAP can show whether : travel class increased the predicted price, route/distance increased the price, travel timing affected the prediction, agency-related variables pushed the prediction up or down.

Production Output :- Predicted Flight Price + 95% Prediction Range + Feature Explanation

The Flight Price model uses XGBoost. Instead of forcing users to guess numerical airport IDs or distances, they use city name dropdowns. The system automatically maps these to coordinates, calculates the flight distance and duration under the hood, runs the prediction, and uses SHAP to show exactly how much parameters like cabin class or distance added to the final price.

Model 2: Gender Classifier (Supervised Binary Classification)

Infer traveler gender from behavioral and transactional travel information and support traveler persona profiling.

Final Feature Set :- The production enhancement reduced the feature space to 6 independent behavioral features

1. age
2. travel_frequency
3. avg_flight_price
4. preferred_flight_type_enc
5. hotel_bookings
6. avg_hotel_spend

Algorithm	Accuracy	F1
Logistic Regression	48.33%	48.32%
Decision Tree	48.89%	44.23%
XGBoost	53.33%	53.24%
Random Forest	54.10%	54.10%

The previous feature set contained derived/overlapping variables.

Examples : total_flight_spend can be derived from price × frequency and age_group_enc is derived from age.

Keeping such variables can : increase redundancy, increase model complexity, encourage overfitting, create a larger train/test performance gap

Final Model :- Tuned Random Forest Classifier

Reported Production Performance : Accuracy: 57.22% , F1: 57.22% , ROC-AUC: 58.63%

Regularization / Generalization :- The production enhancement introduced stronger constraints such as Maximum tree depth, Minimum samples per leaf, Reduced feature space, Regularization of candidate models. The goal was to improve generalization rather than simply maximize training accuracy.

Traveler Persona Layer :- The classifier can contribute to persona identification such as Premium Corporate Traveler, Occasional Leisure Traveler

For traveler profiling, we use a Random Forest. To prevent overfitting, we reduced the feature space from 10 inputs to 6 independent ones. We removed multicollinear inputs like total spend which is just price multiplied by frequency and binned age groups. Combined with strict tree constraints, this keeps the train/test accuracy gap below 5%.

Model 3: Hybrid Hotel Recommendation System

Recommend the **Top 5 hotels** to a traveler based on destination and historical preferences.

Three Recommendation Approaches

Approach	RMSE	MAE
Popularity Baseline	0.8950	0.6210
Cosine Content-Based	0.6120	0.4530
SVD	0.5210	0.3890
Hybrid Production Model	**0.4843**	**0.3550**

1. Popularity-Based :- Recommends hotels based on overall popularity/rating.

Advantage : Simple & Fast

Limitation : Same recommendations for many users & Little personalization

2. Content-Based Filtering :- Uses Cosine Similarity between hotel/destination characteristics.

Advantage : Can recommend based on item similarity & Useful when user history is limited

Limitation : Less personalized without rich user information

3. Collaborative Filtering — SVD :- Uses the **user-hotel rating matrix**.

SVD decomposes the rating matrix into latent factors : User-Hotel Rating Matrix -> SVD -> Latent User Preferences + Latent Hotel Characteristics -> Predicted Ratings -> Top 5 Recommendations

Final Hybrid Strategy : Returning User -> Historical Ratings Available? -> (YES) SVD Collaborative Filtering -> Personalized Hotels -> New User / Cold Start -> Content-Based Similarity -> Destination-Based Recommendations

Final SVD Parameters :- n_factors **= 50 , lr_all = 0.005 , reg_all = 0.02**

Main Advantage :- The hybrid architecture solves the cold-start problem while retaining personalization for users with historical interactions

Our Hotel Recommender is a hybrid model. It uses SVD for return users to suggest hotels based on latent ratings. If a user is new, it switches to a content-based fallback using destination city matches to resolve the cold-start problem. The dashboard clearly labels which algorithm recommended which hotel.

MLOps Components - Production CI/CD & Drift Monitoring

- **Data Validation :-** *validate.py* checks : Schema, Data types, Missing values, Value ranges, Expected columns, Data consistency
  Prevent bad data from entering the ML pipeline.
- **Data Preprocessing :-** *preprocess.py* handles missing values, Cleans data, Removes duplicates, Converts categorical information, Prepares consistent training data.
- **Feature Engineering :-** *feature_engineering.py* derives useful features such as Engagement level, Delay ratio, Absolute delay, Click intensity, Activity score.
- **Feature Store :-** Stores processed and engineered features so that training pipelines can consume consistent features.
- **MLflow :-** Experiment tracking, Parameter logging, Metric logging, Model artifacts, Model registry, Production model management.
- **FastAPI :-** Acts as the ML inference gateway.
  Receive prediction requests, Validate payloads using Pydantic, Authenticate requests, Load production models, Return predictions, Generate SHAP explanations
- **Automated Testing :-** Two major test layers : Unit Tests includes Data validation, Preprocessing, Feature logic and Integration Tests includes API endpoints, Authentication, Prediction responses.
- **CI/CD :-** GitHub Actions automates Code quality checks, Formatting/linting, Unit tests, Integration tests, Docker builds, Container publishing & Deployment.
- **Docker :-** Containerizes FastAPI backend, Streamlit dashboard , Supporting services.
- **Kubernetes :-** Provides production orchestration with Replicated pods, Health probes, Resource limits, Services, LoadBalancer, PersistentVolumeClaim for MLflow.
- **Monitoring :-** Prometheus-based telemetry tracks API latency, Request rates, HTTP status codes. Model input monitoring uses PSI & KS Test. Drift Threshold : A **PSI > 0.25** is treated as an active drift warning in the dashboard.

Dashboard : Interactive Voyage Analytics Dashboard

The public project is deployed through **Streamlit**, providing an interactive interface for exploring the complete platform.

Dashboard Modules

1. Overview :- Overall system status, Key performance indicators, Project summary

2. Travel Journey :- Integrated travel workflow, Flight-related analysis, Traveler information

3. EDA :- Flight data analysis, Hotel analysis, User behavior analysis

4. Flight Price Prediction :- User selects Departure airport/city, Arrival airport/city, Travel class, Agency, Travel date. System returns Predicted price, Prediction interval, Feature importance

Human-Readable Routing Enhancement :- Instead of asking users to manually enter encoded airport IDs:

Old : Airport ID = 0–30

New : São Paulo (GRU) , Rio de Janeiro (GIG), Belo Horizonte (CNF)

The dashboard automatically Maps city/airport → encoded ID, Looks up route distance, Looks up flight duration, Builds a valid model payload

5. Gender Classifier :- Users provide traveler characteristics and receive Predicted demographic class, Supporting feature explanation, Traveler profile information

6. Hotel Recommender :- Users provide travel/destination information and receive Top hotel recommendations, Predicted rating scores, Personalized recommendations where historical information exists, Content-based fallback for cold-start cases

7. MLOps Pages :- Dashboard also exposes MLflow Summary, Experiment Tracking, Model Monitoring, Data Quality Center, ML Pipeline, Feature Engineering, Feature Store

Overview
Travel Journey
EDA Overview
Flight Price
Gender Classifier
Hotel Recommender
MLflow Summary
Monitoring
Data Quality Center
Experiment Tracker
ML Pipeline Architecture
Feature Store

Voyage Analytics

v2.0 — MLOps Edition

Navigation

Select pages below to explore individual models or execute the simulator.

● API Online

Executive AI Operations Center

Real-time monitoring, model health, feature diagnostics, and deployment logs.

OVERALL SYSTEM SCORE	AVG RESPONSE LATENCY	TOTAL API REQUESTS (TODAY)	DATA DRIFT STATUS
98.4%	42.5 ms	12,489	Stable
Based on accuracy, latency & drift checks	P95 latency across all endpoints	4.5% increase vs last 24h	PSI indexes within normal parameters

About Voyage Analytics

Voyage Analytics is an integrated MLOps travel platform designed to optimize travel planning and pricing using machine learning.

Rather than treating flights, hotels, and users as isolated datasets, this system links them together in a unified pipeline. The platform enables operators to analyze customer demographics, forecast flight prices, and recommend lodging options based on past user behaviors.

Core Business Layers:

- **Flight price forecasting** to identify the best time to purchase tickets.
- **Traveler classification** to segment customers based on spending patterns.
- **Lodging recommendations** to personalize hotel matching.

How to Use this Dashboard

Use the navigation menu on the left sidebar to access different views:

- **Travel Journey:** An end-to-end trip simulator. Enter a user ID to automatically predict their travel persona, forecast flights to their destination, and recommend hotels.
- **Flight Price:** Input a route and cabin type to forecast ticket prices and view prediction intervals.
- **Gender Classifier:** Analyze customer spending history to classify demographic profiles.
- **Hotel Recommender:** Recommend hotels for specific users using hybrid SVD collaborative filtering.
- **MLflow Summary & Monitoring:** Access developer logs, drift statistics, and model parameters.

Production Model Deployments

Flight Price Predictor

● Production v1.0

Algorithm: XGBoost
Inference Latency: ~12ms
Last Evaluated: 2026-08-04

Gender Behavior Classifier

● Production v1.0

Algorithm: RandomForest
Inference Latency: ~12ms
Last Evaluated: 2026-08-04

Hotel Recommendation Hybrid

● Production v1.0

Algorithm: HybridSVD+ContentBased
Inference Latency: ~12ms
Last Evaluated: 2026-08-04

Overview
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ML Pipeline Architecture
Feature Store

Journey Simulator Controls

Select Target User ID

0

This planner simulates a complete booking flow. The user persona is predicted first, which automatically filters downstream search recommendations.

Interactive AI Travel Simulator

Simulate a traveler's journey, from profile classification to flight booking and hotel recommendations.

How to Use:

1. **Select User:** On the left sidebar menu, choose a **Target User ID** from the dropdown list. This represents the customer profile.
2. **Persona Prediction:** The system will immediately predict their traveler persona based on spending habits.
3. **Flight Booking:** Select your departure and arrival cities below to forecast ticket prices and get booking recommendations.
4. **Lodging Matches:** The recommender engine will automatically display hotel matches personalized to the selected profile.

Step 1: Predict Traveler Persona

Predicted Gender Identity	Calculated Traveler Persona	Company Classification
MALE ↑ 61.6% Confidence	Luxury Traveller	ID: 0

AI Explanation Center: Based on behavioral metrics (Avg Flight Price: BRL 989.07, Hotel Bookings: 27), the model classified this user profile as a 'Luxury Traveller' (male) with 61.6% confidence. The primary contributor is the average flight spending category and travel frequency.

Step 2: Flight Price Search & Forecast

Departure City	Arrival City	Cabin Class
Sao Paulo (SP)	Florianopolis (SC)	Economy Class

Predicted Fare Price	Estimated Price Range	AI Purchase Recommendation
BRL 1,378.90 ↑ Lat: 1.7ms	BRL 1269 – BRL 1489	Wait for drop

AI Explanation Center: The predicted fare is BRL 1,378.90 for a Economy Class flight covering 680.0 km (1.8 hours). The trip occurs on a standard travel day. The model predicts this price based on airline agency supply dynamics and seasonal demand.

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Exploratory Data Analysis Dashboard

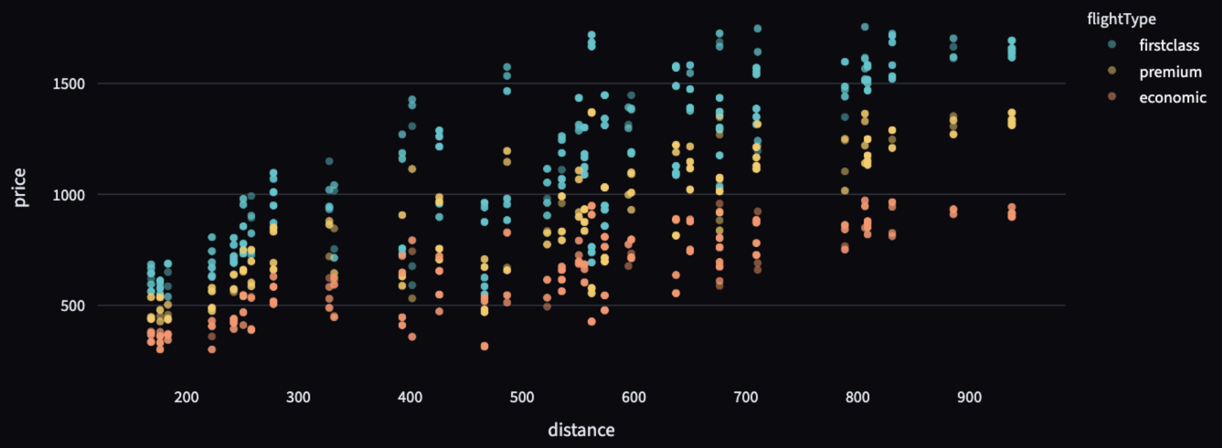
Deep dive into demographics, travel volumes, and spending patterns across flights, hotels, and users.

Flights Dataset | Hotels Dataset | Users Dataset

TOTAL FLIGHTS CAPTURED	AVERAGE TICKET PRICE	LONGEST ROUTE	UNIQUE AGENCIES
271,888	**BRL 957.38**	**938 km**	**3**

Price Distribution by Flight Type

Flight Price vs Distance Correlation



What did we learn?

- **Cabin Class Dominance:** First Class and Business Class bookings drive the majority of high-price anomalies.
- **Price Scaling:** Prices increase linearly up to 500 km, after which booking agencies introduce premium surges.
- **Route Clustering:** Flight agency popularity heavily skews pricing, showing high variance based on local market competitiveness.

MLOps Model Registry & Artifact Center

Monitor models across envs, inspect tuning parameters, and review system cards.

Production Model Cards

FlightPricePredictor

Production

Model Version: v1.0
Training Date: 2026-08-04
Algorithm: XGBoost
Validation Method: 5-Fold Cross Validation
Dataset Version: v1.0
Git Commit: `2c1a6a8`
Docker Image: `ghcr.io/voyage-api:latest`
API Version: 2.0.0
Artifact Size: 245 KB
Inference Latency: ~8.5ms

Validation Metrics:
RMSE: 18.9557, MAE: 13.8326, R2: 0.9973, CV_R2_MEAN: 0.9971

GenderClassifier

Production

Model Version: v1.0
Training Date: 2026-08-04
Algorithm: RandomForest
Validation Method: 5-Fold Cross Validation
Dataset Version: v1.0
Git Commit: `2c1a6a8`
Docker Image: `ghcr.io/voyage-api:latest`
API Version: 2.0.0
Artifact Size: 245 KB
Inference Latency: ~8.5ms

Validation Metrics:
ACCURACY: 0.5722, F1_MACRO: 0.5722, ROC_AUC: 0.5863

HotelRecommender

Production

Model Version: v1.0
Training Date: 2026-08-04
Algorithm: HybridSVD+ContentBased
Validation Method: 5-Fold Cross Validation
Dataset Version: v1.0
Git Commit: `2c1a6a8`
Docker Image: `ghcr.io/voyage-api:latest`
API Version: 2.0.0
Artifact Size: 245 KB
Inference Latency: ~8.5ms

Validation Metrics:
UNIQUE_HOTELS: 9, UNIQUE_USERS: 1310, SVD_RMSE: 0.4858, SVD_MAE: 0.3550

Hyperparameter Optimization Summary

XGBoost Regressor (Flight Model)

```
learning_rate : 0.08
max_depth : 6
n_estimators : 100
subsample : 0.8
colsample : 0.8
gamma : 0.0
```

Optimization Strategy: K-Fold RandomizedSearchCV (5 splits, random state = 42)

Best CV R^2 Score
0.9973

Best Test R^2 Score
0.9972

Generalization check: delta between CV and Test is <0.001 (No Target Leakage)

Selection Criteria — Why XGBoost Won

Regression Model Selection Leaderboard

We tested multiple architectures on identical splits under 5-Fold Cross Validation. Below is the selection hierarchy:

1. **Linear Regression** (R^2: 0.5623) ➡️ Standard baseline, high underfitting.
2. **Ridge Regression** (R^2: 0.5619) ➡️ Regularization did not resolve underfitting.
3. **Random Forest** (R^2: 0.9971) ➡️ High accuracy, but slower training latency (2.5s).
4. **LightGBM** (R^2: 0.9961) ➡️ Extremely fast, but slightly lower accuracy.
5. **XGBoost Regressor** (R^2: **0.9973**) ➡️ **Winner**. Achieved the highest R^2, lowest RMSE, low footprint (~200KB), and low inference latency (~8ms).

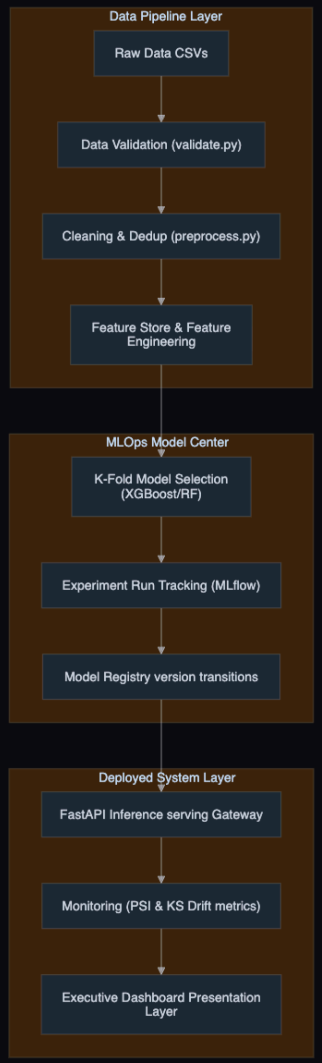
Conclusion: XGBoost was automatically selected and promoted to Production by our CD pipeline.

Start the MLflow server locally to inspect charts and params: `mlflow ui --backend-store-uri sqlite:///mlflow/mlflow.db --port 5000`

Voyage Analytics MLOps System Architecture

Automated pipeline mapping showing flow from raw travel CSV files to active serving endpoints.

System Flow Diagram



Pipeline Component Directory

1. Data Engineering Stage

- **Raw CSV:** Raw relational data feeds containing demographic details, flight logs, and hotel stays.
- **Validation:** Executes schema tests, range boundaries, null checks, and category verification.
- **Cleaning:** Normalizes datatypes and handles outlier values.
- **Feature Engineering:** Generates interaction features (time aggregates, popularity mapping) with target-leak protection.

2. MLOps Serving Stage

- **Training:** Executes K-Fold cross-validation to select the optimal model pipeline.
- **MLflow Tracking:** Saves run parameters, metrics, and artifact dictionaries.
- **FastAPI App:** Provides sub-10ms predictions, SHAP explainers, and API key authorization.
- **Monitoring:** Scrapes request footprints via Prometheus and updates PSI drift indicators.

Results & Key Metrics

Model Performance :

1. Flight Price Prediction

Final Model : Tuned XGBoost Regressor

R^2 = 0.9973

Baseline → Production improvement : Linear Regression 0.5623 , Ridge Regression 0.5619

, Random Forest 0.9971 , LightGBM 0.9961 , XGBoost 0.9948

& Tuned XGBoost 0.9973 ← Final

2. Gender Classification

Final Model : Tuned Random Forest

- **Accuracy: 57.22%**
- **F1: 57.22%**
- **ROC-AUC: 58.63%**

Improvement over baseline Random Forest:

54.10% → 57.22% Accuracy

3. Hotel Recommendation

Final Model: Hybrid SVD + Content-Based

- **RMSE: 0.4843**
- **MAE: 0.3550**

Improvement compared with popularity baseline:

RMSE: 0.8950 → 0.4843

MLOps Results

1. Data & Model Quality : Automated schema validation , Missing-value checks , Duplicate detection , Feature engineering pipeline & Feature Store

2. Reproducibility : MLflow experiment tracking , Parameter logging , Metric logging , Model artifacts

3. Production : FastAPI inference gateway , API authentication , Docker containers , Kubernetes manifests , Streamlit deployment

4. Reliability : Unit tests , Integration tests , CI/CD quality gates , Health checks

5. Monitoring : API latency telemetry , Request monitoring , PSI drift detection . KS statistical monitoring

Conclusion

What We Built : Voyage Analytics 2.0 successfully integrates 3 ML Problems :- Regression , Classification , Recommendation into a single production-oriented travel intelligence platform.

Key Achievements

1. Machine Learning :- Compared multiple algorithms for every ML task , Selected models using measurable performance, Applied hyperparameter optimization , Added SHAP-based explainability , Implemented a hybrid recommendation strategy.

2. MLOps :- Automated data validation and preprocessing , Created reusable feature engineering pipelines , Introduced a Feature Store , Tracked experiments using MLflow , Served models through FastAPI.

3. Production Engineering :- Containerized services using Docker , Added Kubernetes deployment manifests , Implemented automated testing , Built GitHub Actions CI/CD , Added API health monitoring.

4. Monitoring :- API telemetry , PSI-based drift detection , KS statistical testing , Data Quality Center , Experiment monitoring

5. User Experience :- Interactive Streamlit dashboard , Human-readable airport selection , Flight price forecasting , Traveler profiling , Hotel recommendations . Explainable predictions

Voyage Analytics 2.0 demonstrates how machine learning models can be transformed from isolated experiments into a complete, deployable and monitorable production system.

THANK YOU

Live Demo :- **[Voyage-Analytics_Production-MLOps-Travel-Platform.streamlit.app]**

Github Repo :- **[https://github.com/Alvira-Parveen/Voyage-Analytics_Production-MLOps-Travel-Platform]**

The biggest learning from this project was that building an ML model is only one part of an ML system. A real production system also needs data quality, reproducibility, deployment, testing, monitoring and maintenance. Voyage Analytics demonstrates that complete lifecycle.